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MILANO 1863

EFFECT OF REAR WING ATTACHMENT ON TESLA MODEL S AERODYNAMIC PERFORMANCE

Red bull te da alaaas	
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057233 – AERODYNAMICS OF TRANSPORT VEHICLES

062181 – VEHICLE AERODYNAMICS

PROF. ZANOTTI – PROF. SCHITO

ACADEMIC YEAR 2024/2025



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1. INTRODUCTION



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2. OBJECTIVES

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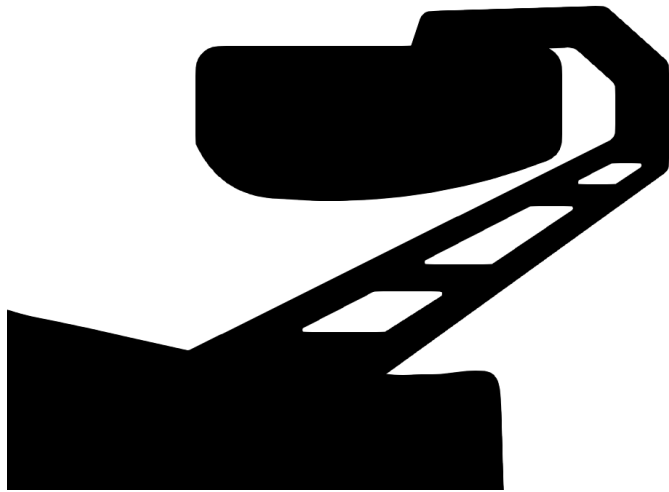
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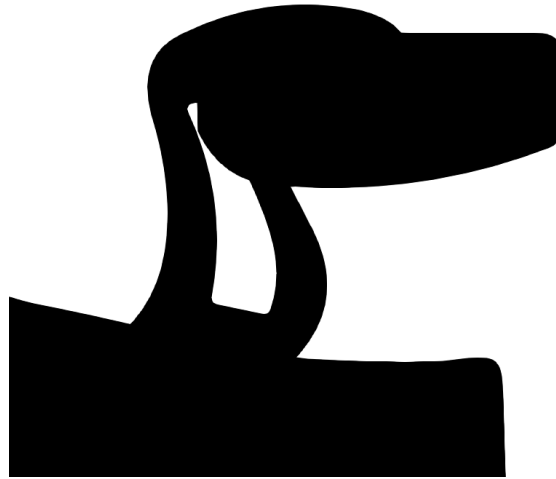
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2. OBJECTIVES

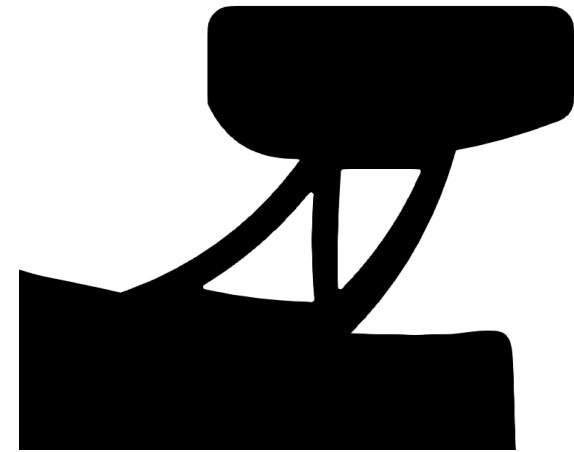
DIFFERENT CASES



Case 1: Back



Case 2: Front



Case 3: Below



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3. METHODOLOGY

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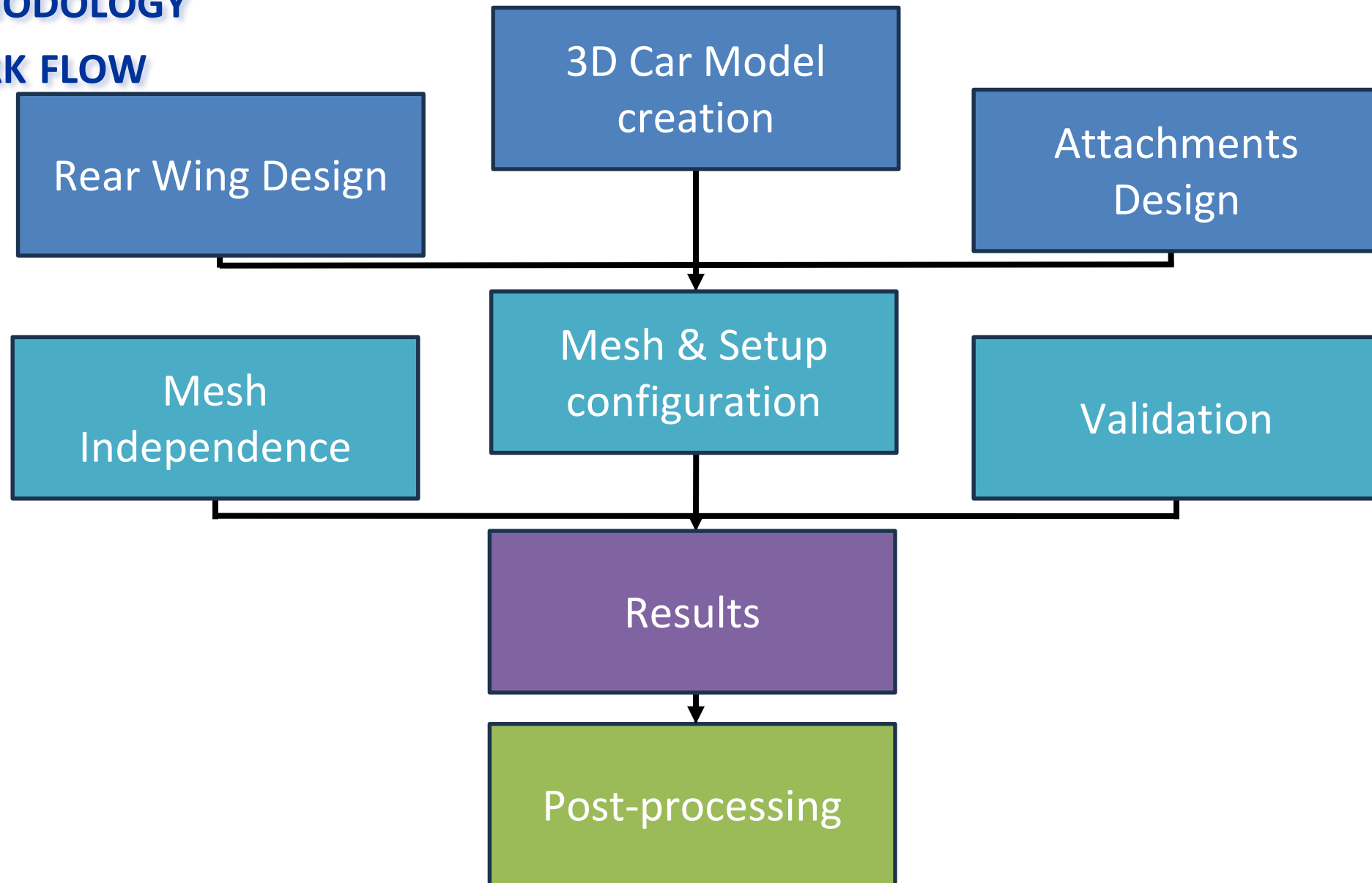
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3. METHODOLOGY

WORK FLOW



3. METHODOLOGY

MODEL CREATION



STL  **CAT PART**

- Watertight
- Wheels added

Rear wing

- NACA 4412
- Selig 1223

3. METHODOLOGY

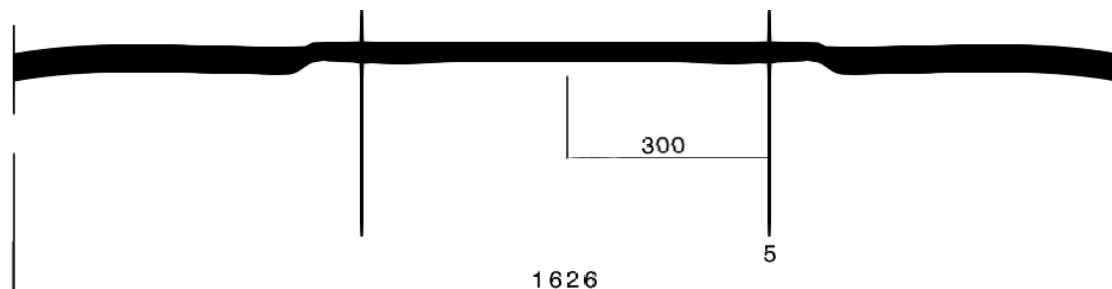
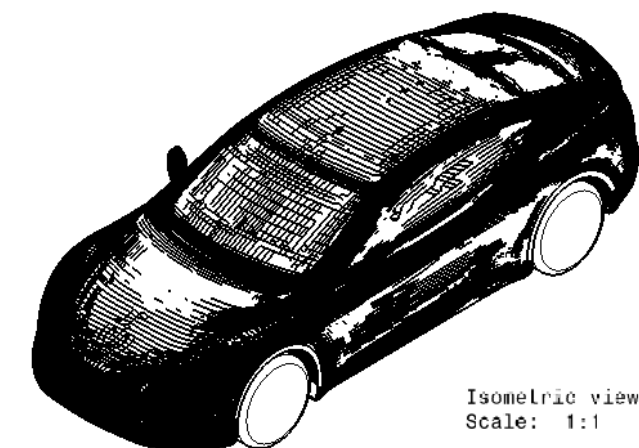
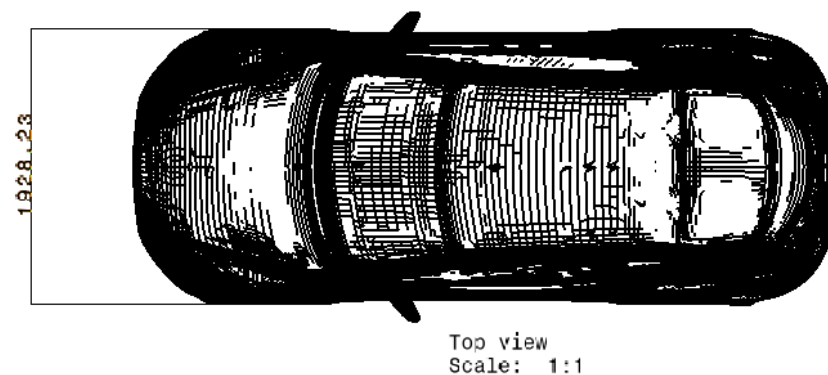
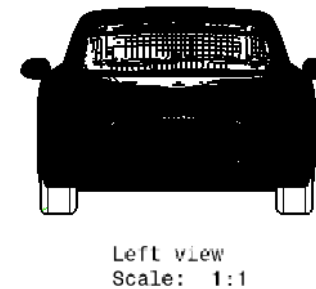
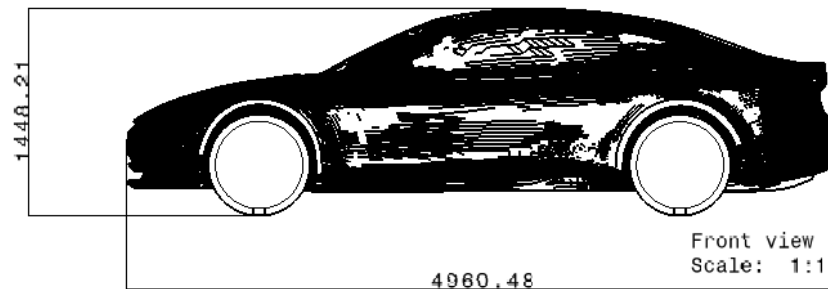
MODEL DIMENSIONS

Car

- $L = 4.96 \text{ m}$
- $W = 1.92 \text{ m}$
- $H = 1.44 \text{ m}$

Rear wing

- $b = 1.626 \text{ m}$
- $c = 0.25 \text{ m}$
- $w = 5 \text{ mm}$

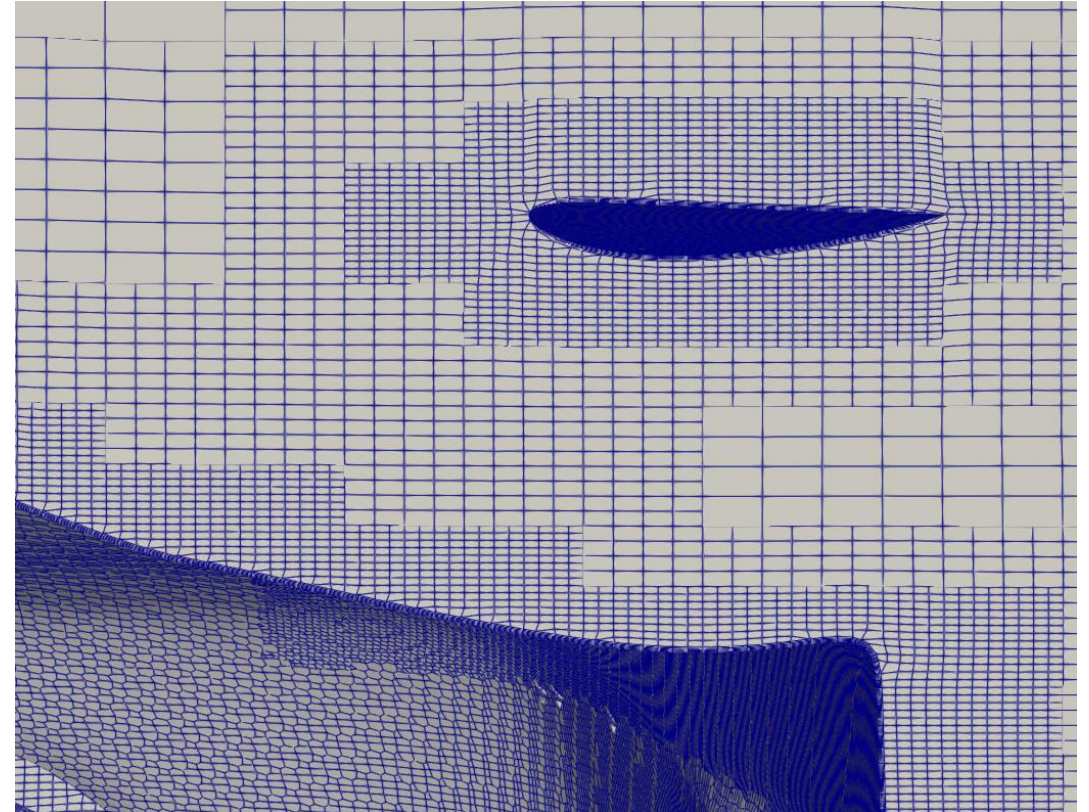
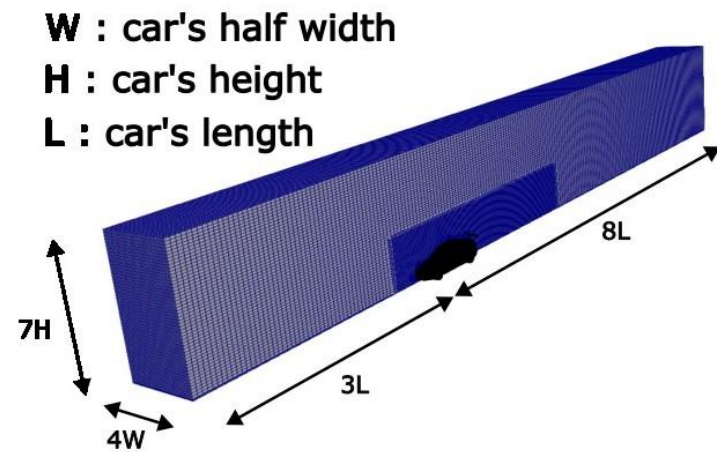


3. METHODOLOGY

MESHING

SnappyHexMesh

- Refinement region around the car
- Refinement box around the rear wing and attachments

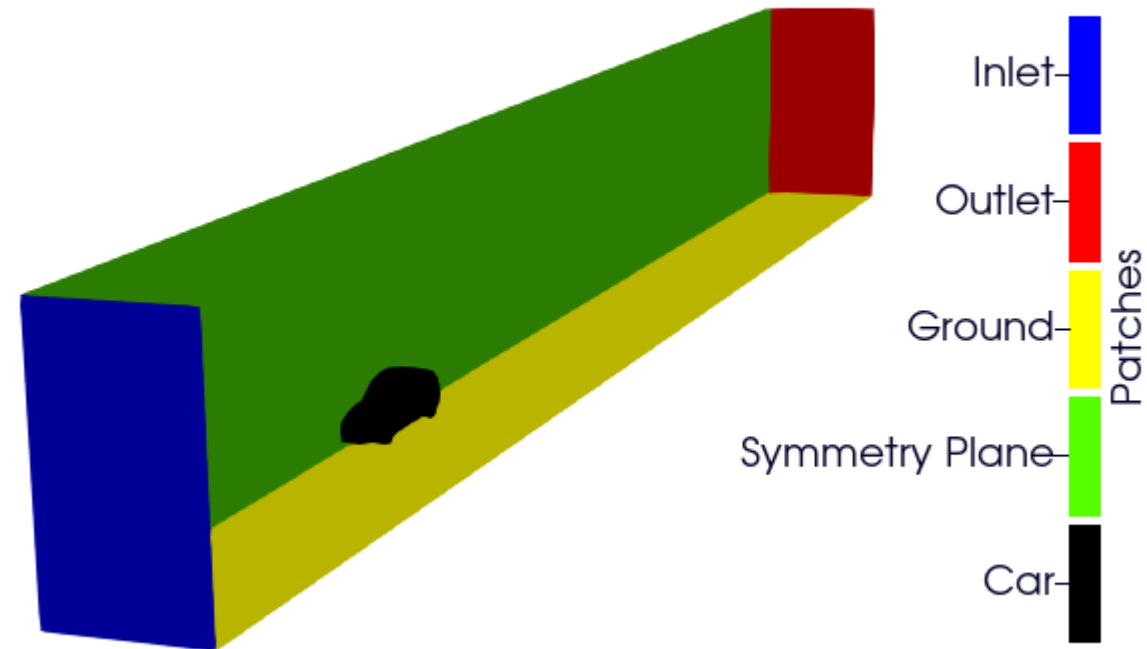


3. METHODOLOGY

BOUNDARY CONDITIONS

Boundary conditions:

- **Inlet** : Velocity inlet
- **Outlet** : Pressure outlet
- **Ground** : No slip
- **Symmetry Plane**
- **Car** : No slip
- **Upper and Lateral planes** : Slip

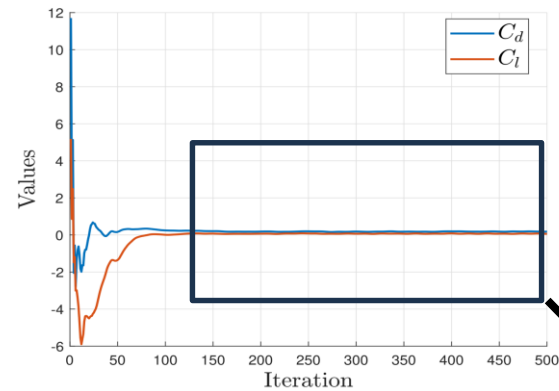


3. METHODOLOGY

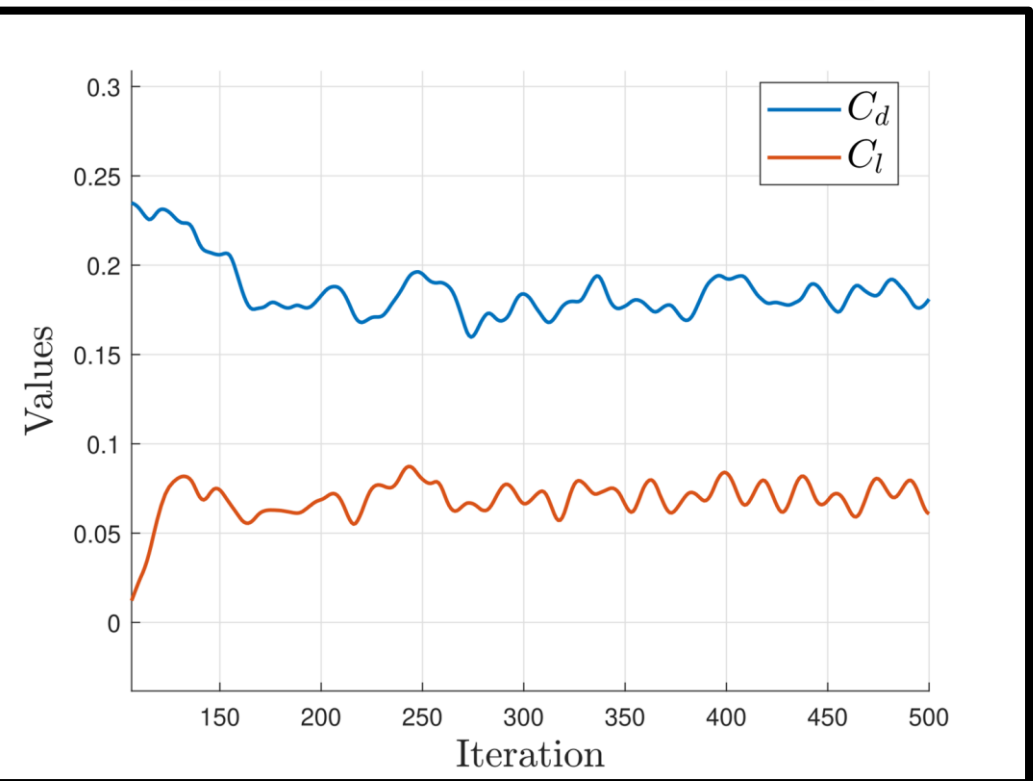
MESH INDEPENDENCE AND RESIDUALS

Basecase

Coarse	3.2
Medium	5.8
Fine	8.4



Aerodynamic coefficients evolution

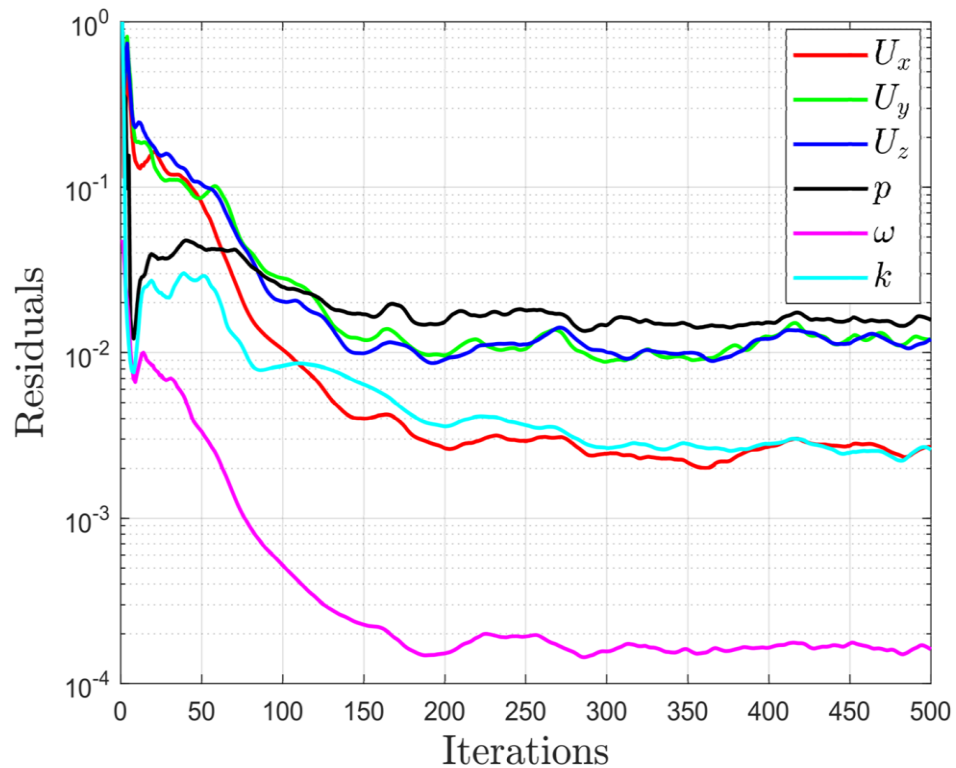


Coarse vs Medium Medium vs Fine

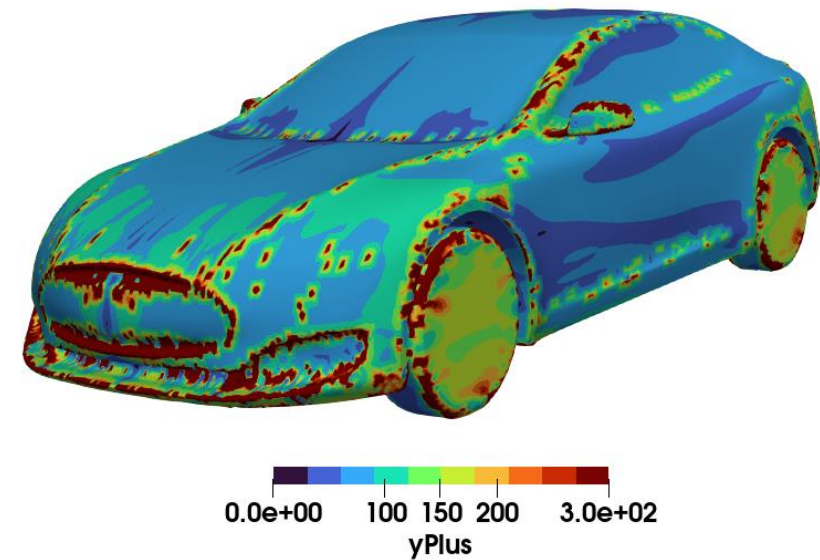
C_L [%]	9.17	7.05
C_D [%]	3.21	6.95

3. METHODOLOGY

MESH INDEPENDENCE AND RESIDUALS



Residuals



Y+ distribution



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4. RESULTS

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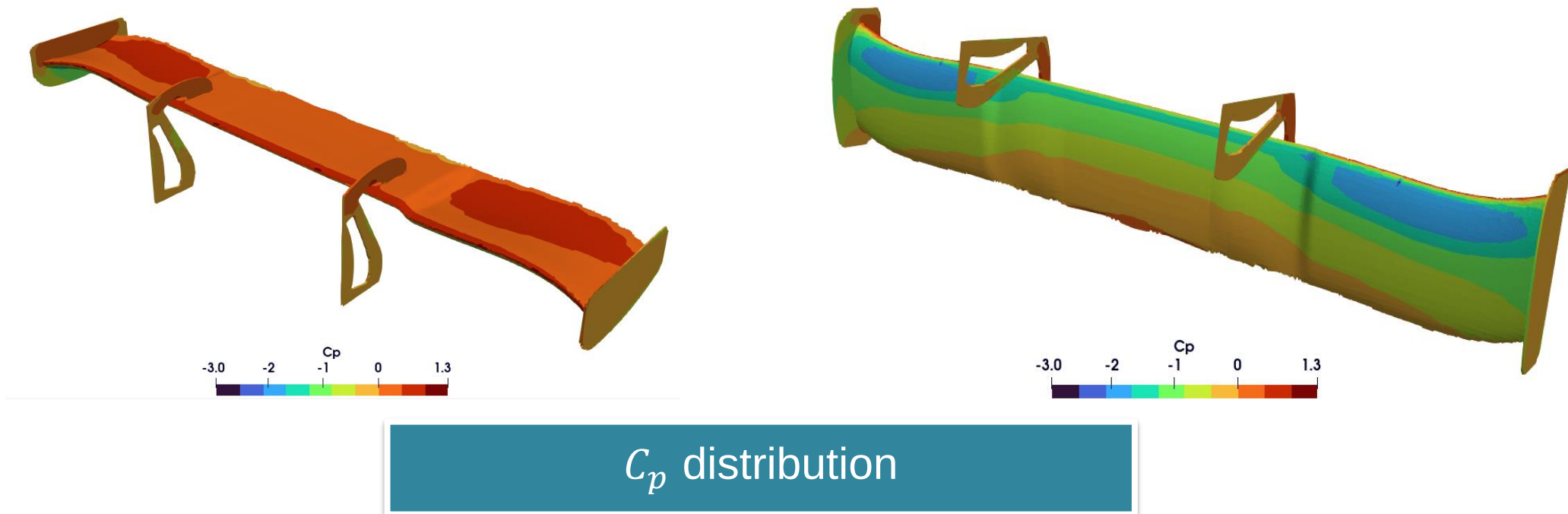
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4. RESULTS

LIFT AND DRAG COEFFICIENTS

	Base	Front	Below	Back
C_L	-0.1120	-0.3506	-0.3513	-0.3268
C_D	0.3640	0.3910	0.3903	0.3642
$ E $	0.3077	0.8966	0.9000	0.8973



4. RESULTS

LIFT AND DRAG COEFFICIENTS

Basic dynamic model to assess cornering performance:

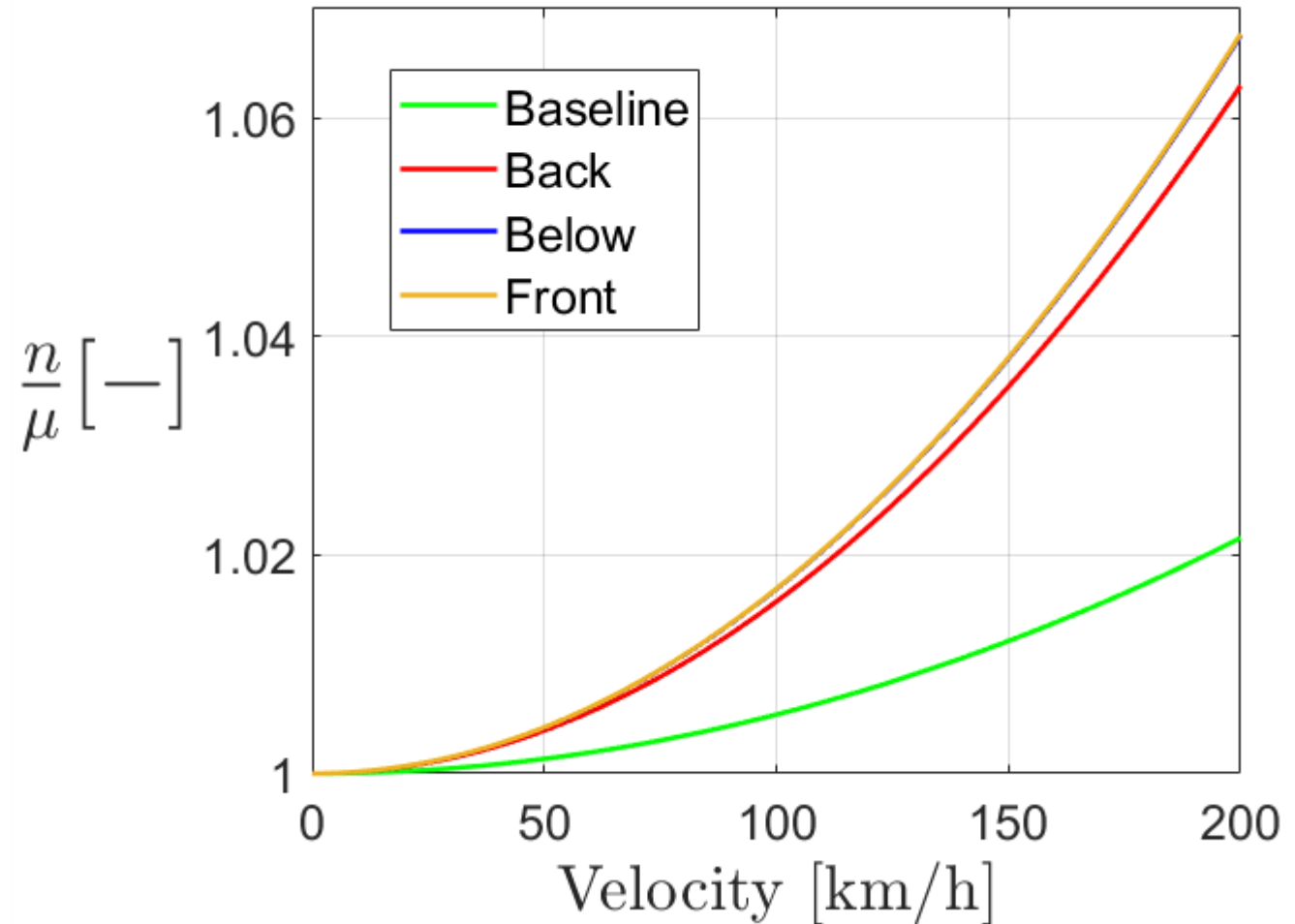
$$nW = \mu(W - L)$$

Standard atmospheric conditions: $\rho = 1.225 \text{ kg/m}^3$

Reference surface: 2.2 m^2

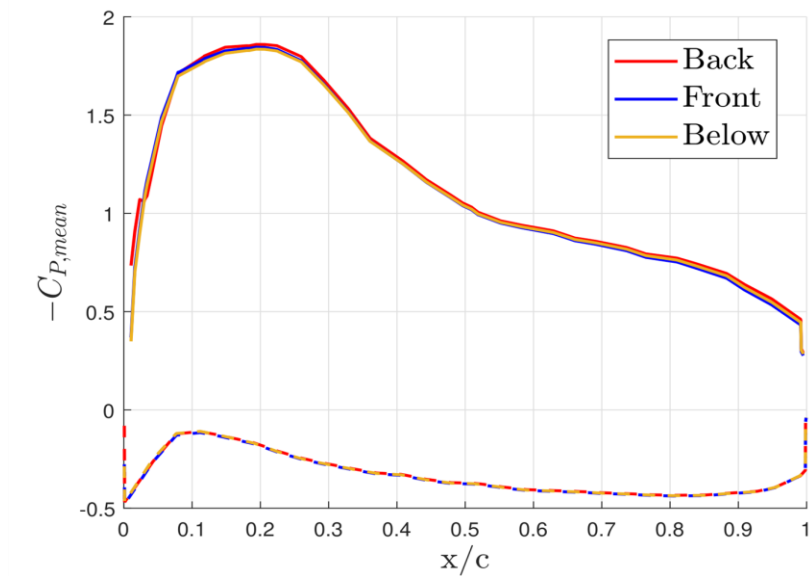
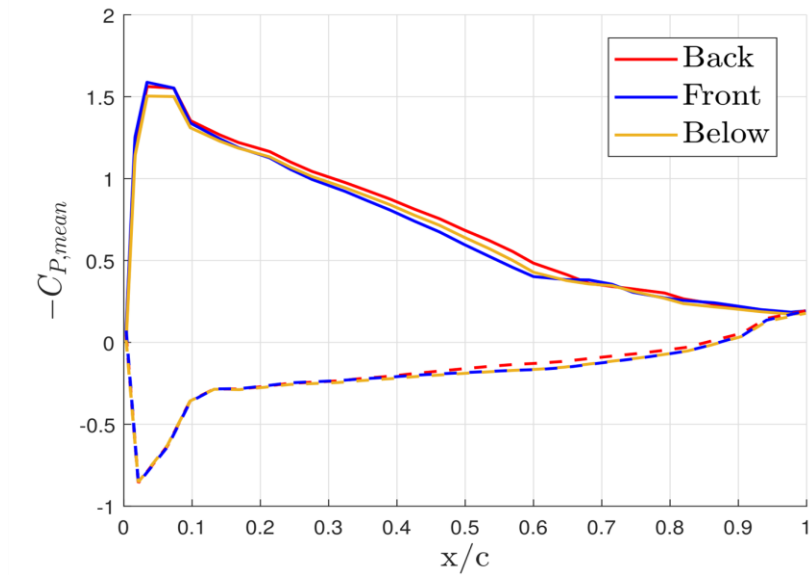
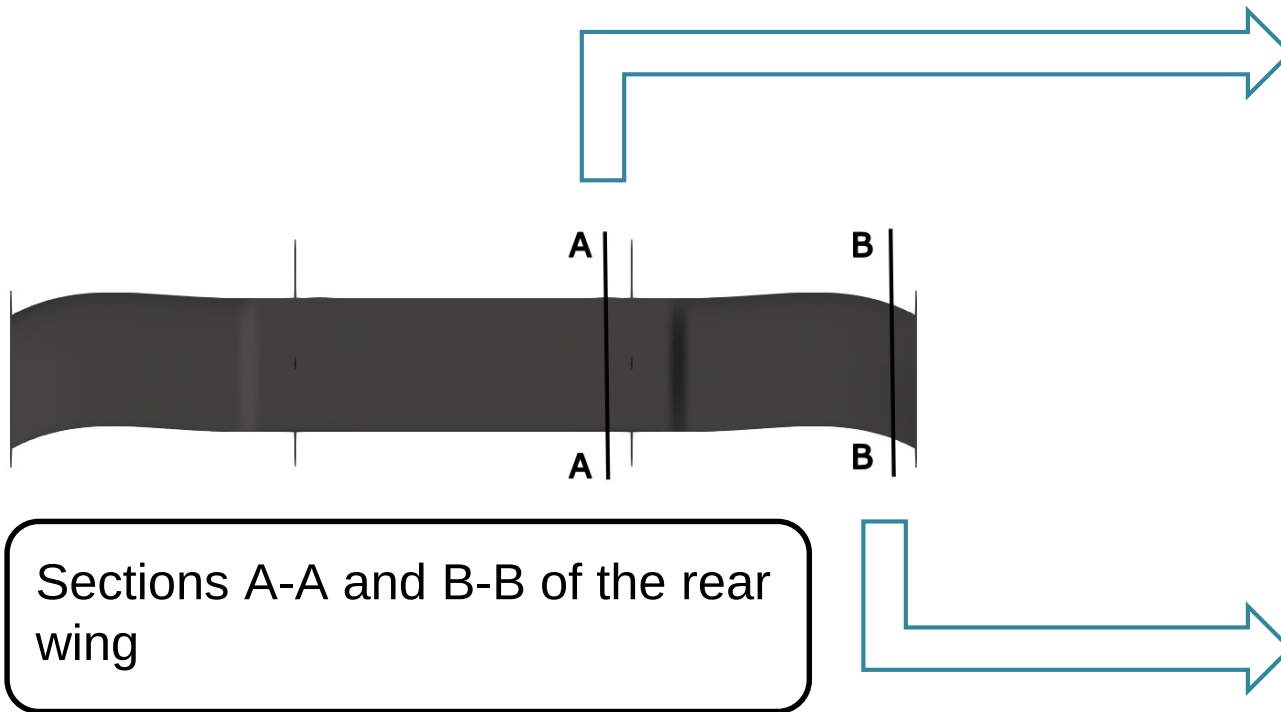
Car mass = 2200 kg

Load factor to friction coefficient ratio



4. RESULTS

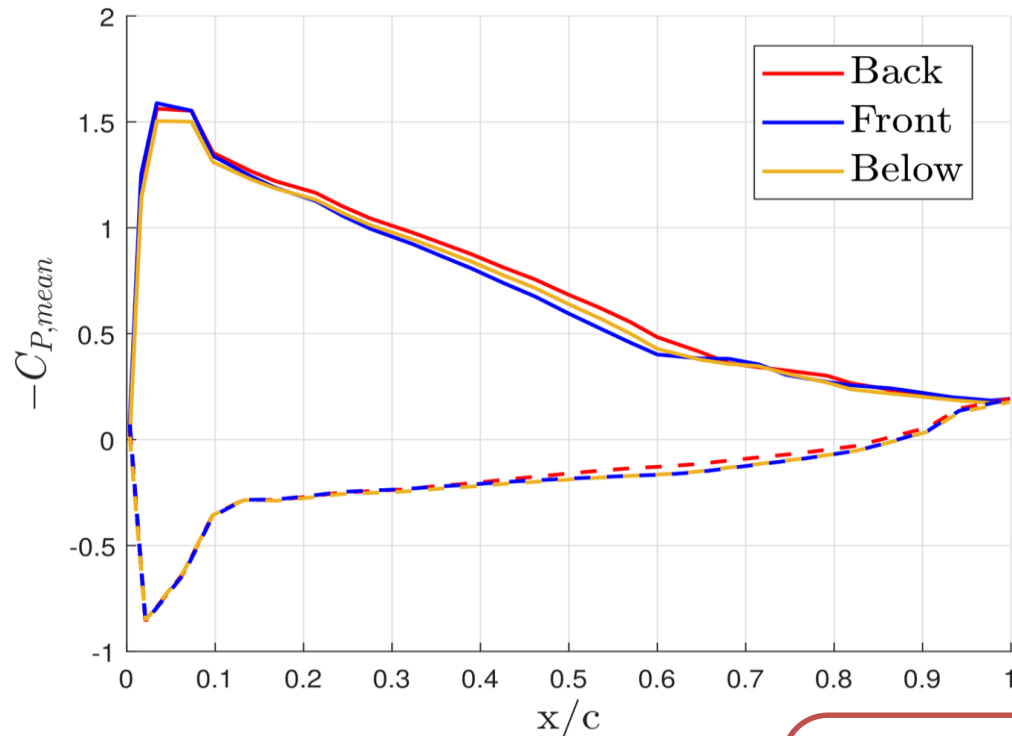
PRESSURE: DISTRIBUTION AT THE REAR WING



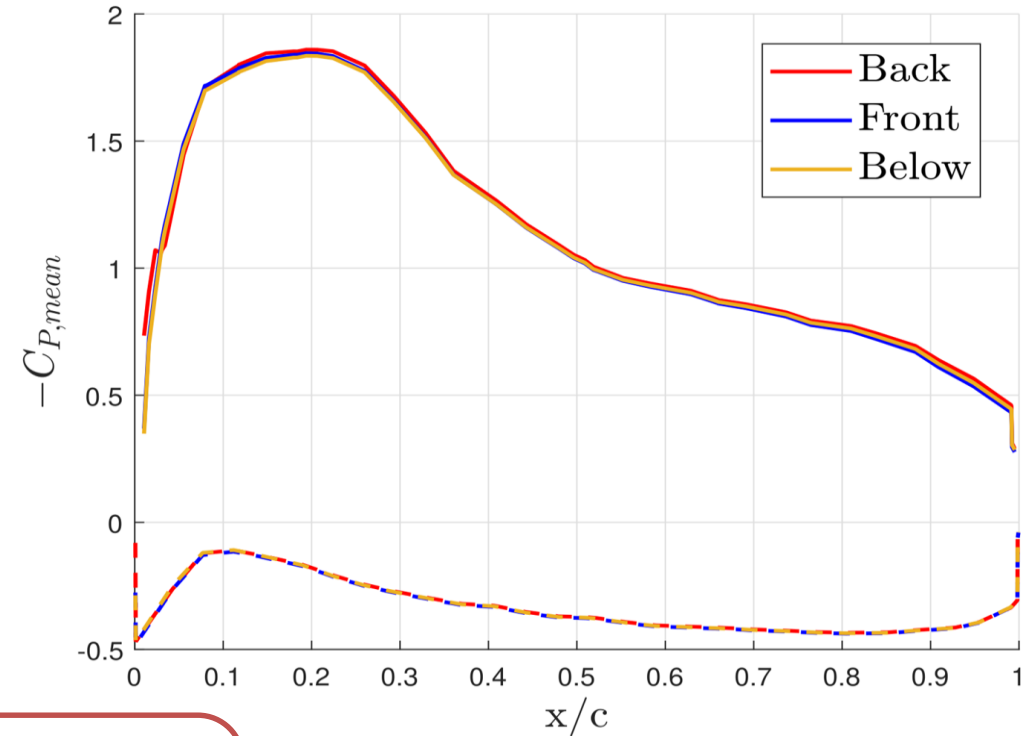
4. RESULTS

PRESSURE: $C_{p,mean}$ DISTRIBUTION AT THE REAR WING

A-A section



B-B section



3D nature of the flow

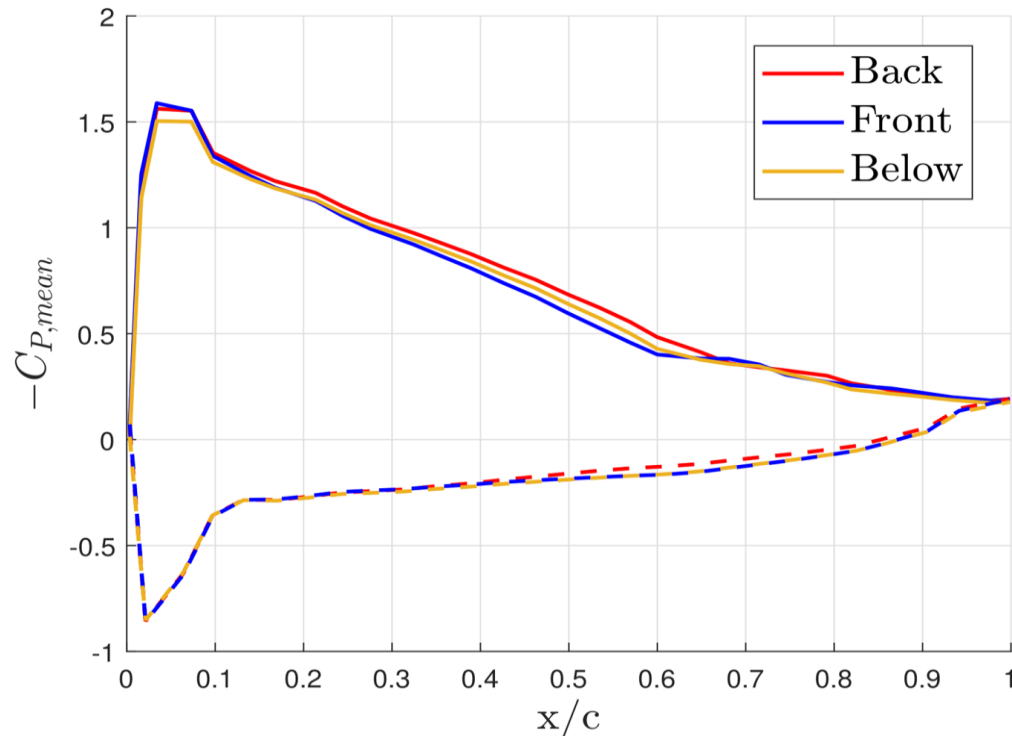
- "Dirty" air from the car
- Spanwise curvature
- Endplates presence

Complex pressure distribution

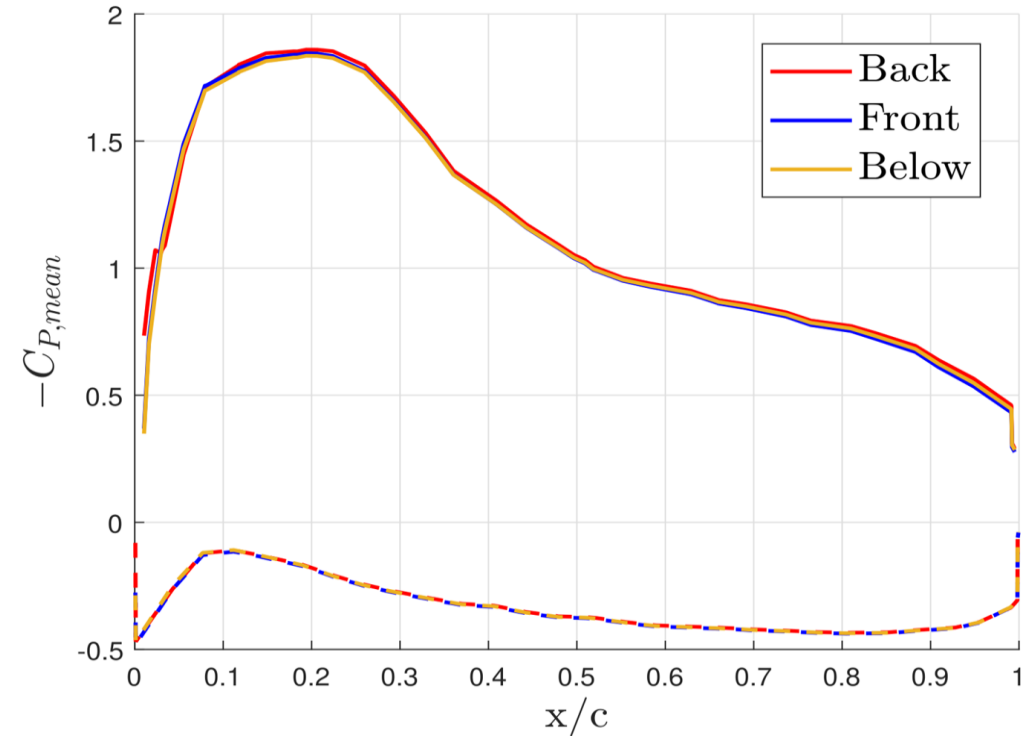
4. RESULTS

PRESSURE: $C_{p,mean}$ DISTRIBUTION AT THE REAR WING

A-A section



B-B section

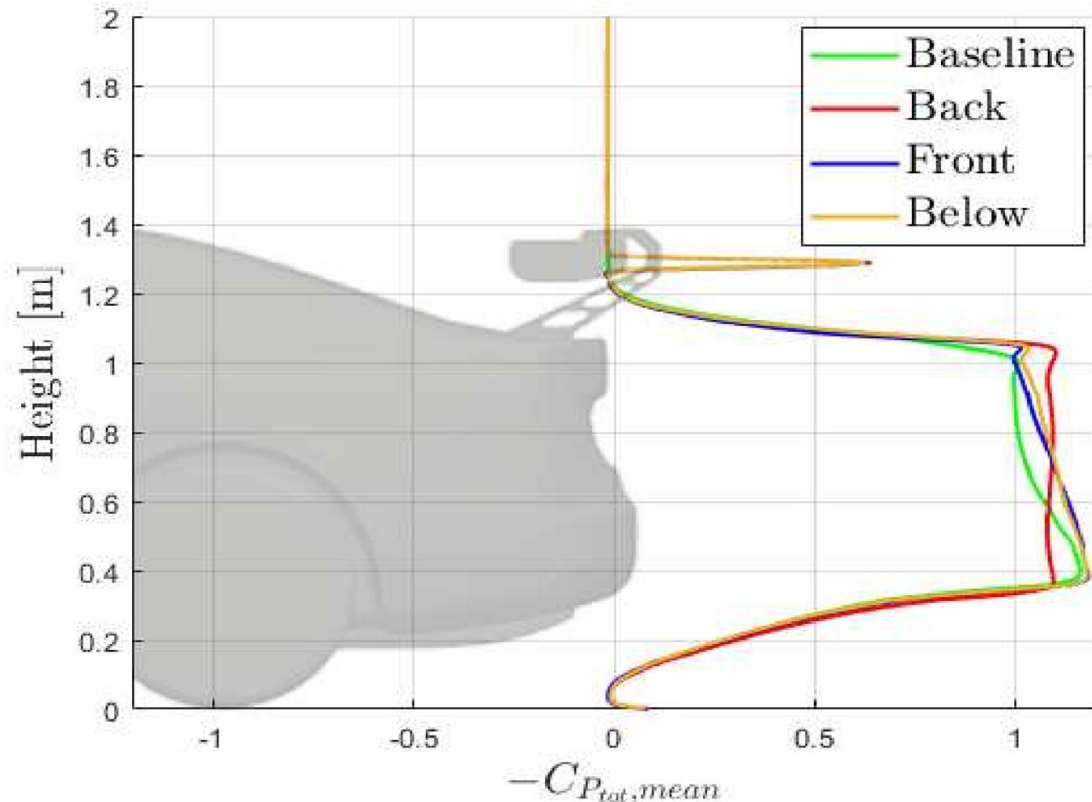


Differences between attachments

Unaffected by different attachments

4. RESULTS

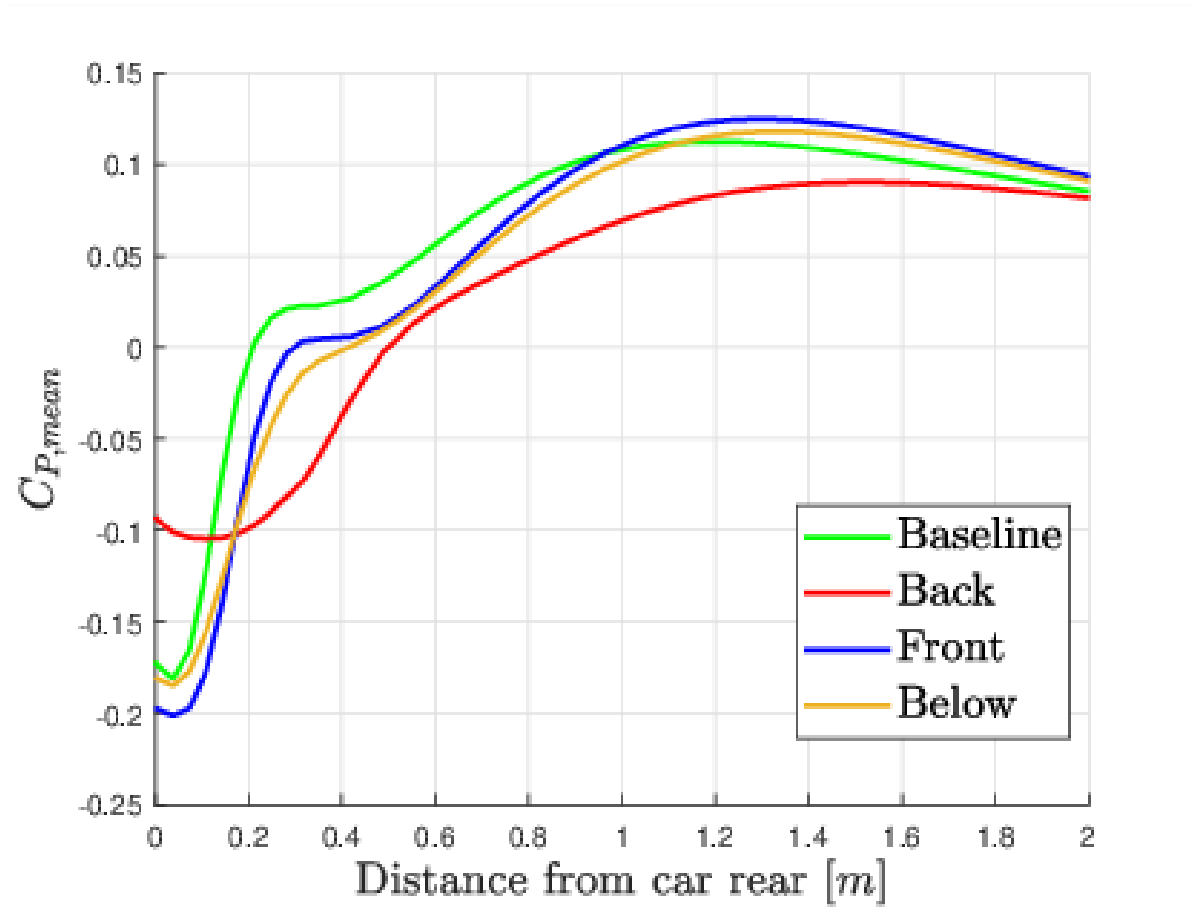
PRESSURE: $C_{p,tot,mean}$ DISTRIBUTION AT THE REAR OF THE CAR



- Rear wing wake: identical for each case
 - Rear of the car wake: interference of attachments
- ↓
- *Back*: uniform wake
 - *Front* and *Below*: non-uniform wake

4. RESULTS

PRESSURE: $C_{p,mean}$ DISTRIBUTION AT THE REAR OF THE CAR



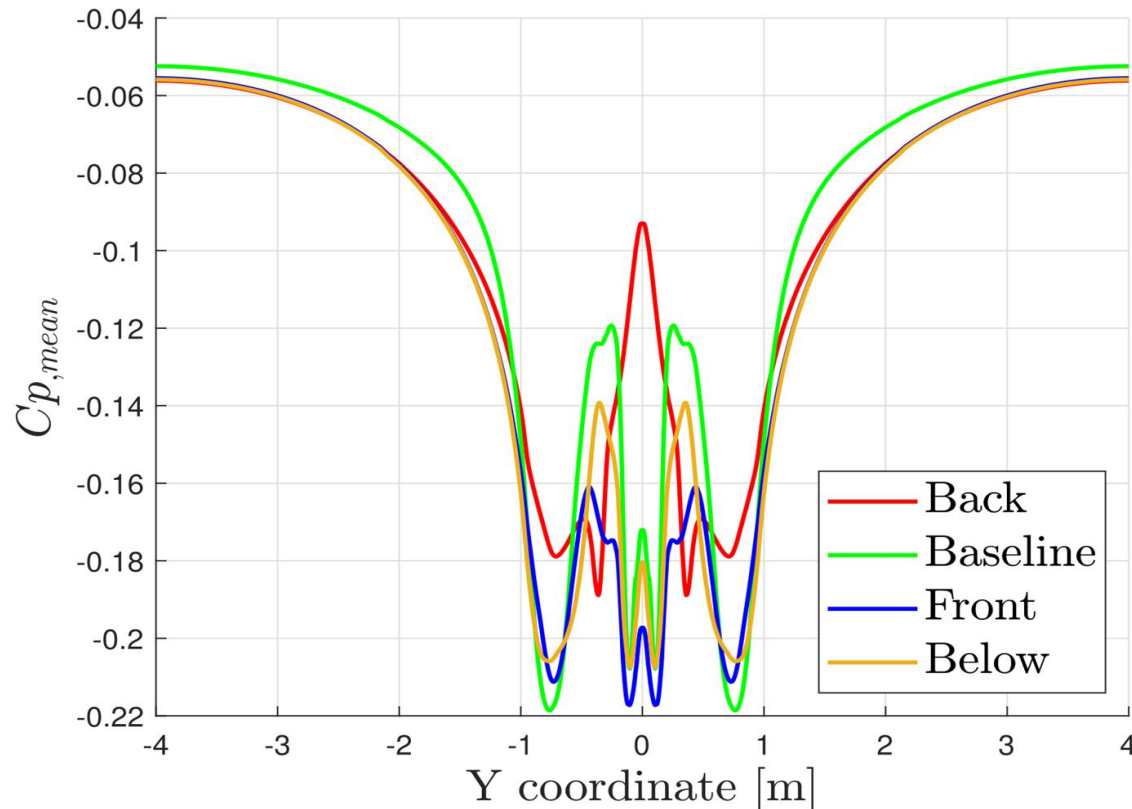
- Cut at a distance of 0.5 m from the ground



- *Baseline*, *Front* and *Below*: similar pressure close to the car
- *Back*: higher base pressure = less pressure drag

4. RESULTS

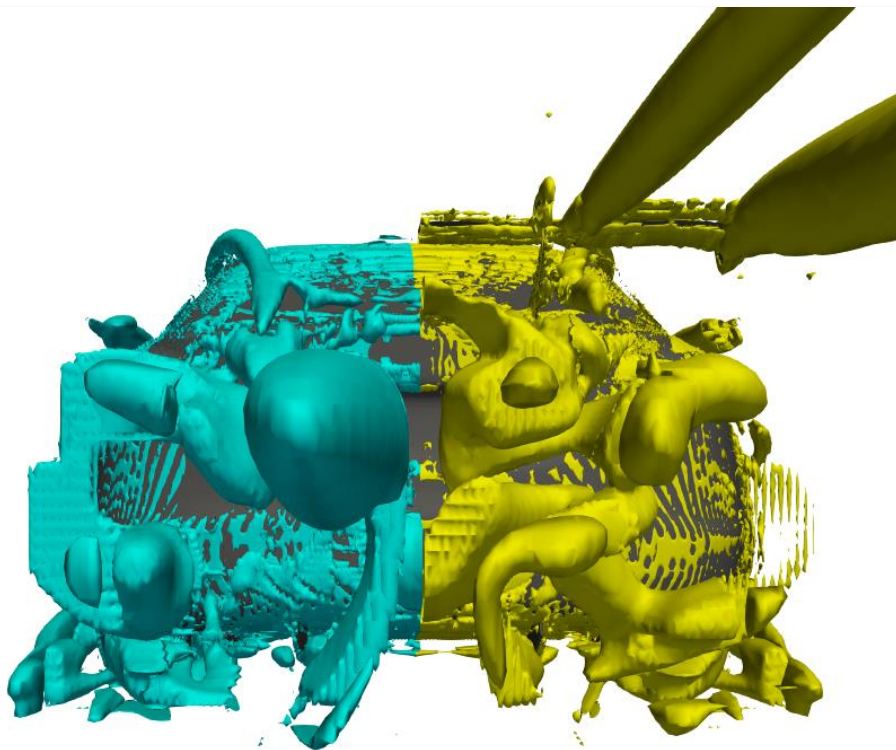
PRESSURE: $C_{p,mean}$ DISTRIBUTION AT THE REAR OF THE CAR



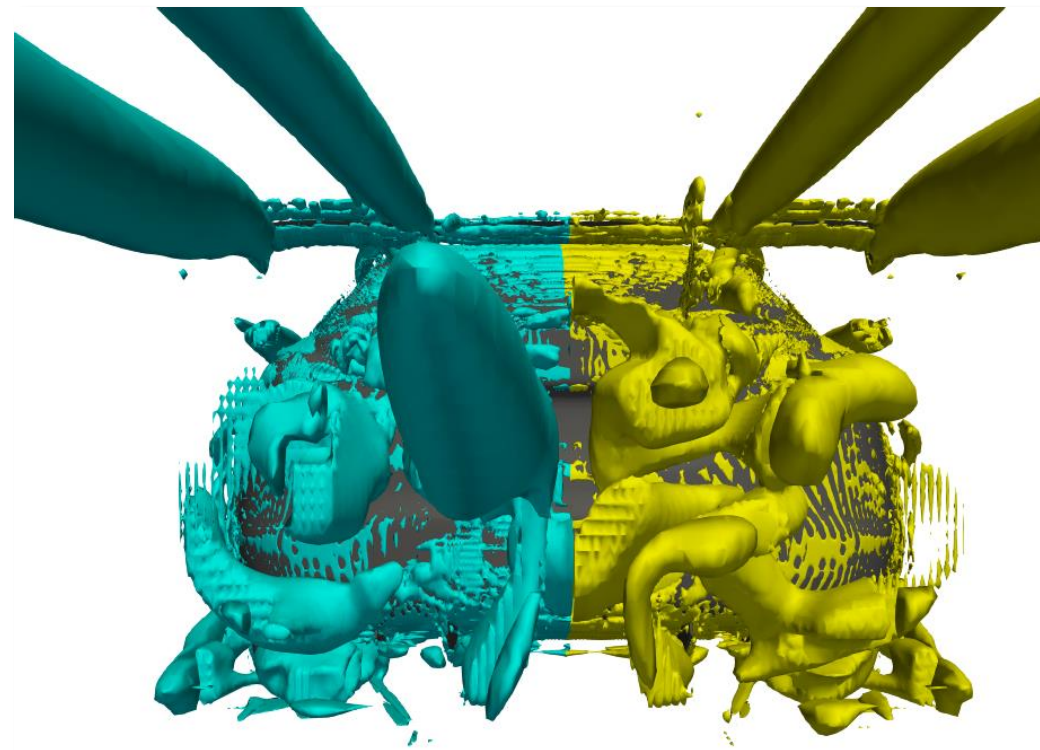
- Transverse evolution
 - Y coordinate is 0 at the symmetry plane
- ↓
- **Back**: higher mean value = lower pressure drag
 - **Front** and **Below**: lower mean value = higher pressure drag

4. RESULTS

Q CRITERION CONTOURS



Baseline (blue) vs Back (yellow)



Below (blue) vs Back (yellow)

4. RESULTS

RATIONALISATION OF RESULTS

Rear wing

- More downforce: induced drag
- More wet surface: friction drag

Interference

- Rear wing and attachment affect wake
- Beneficial for back, detrimental for others
- Complex pressure field on wing

Comparison

- Back: reduced pressure drag, increased friction drag
- Others: increase of all drag sources





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5. CONCLUSIONS

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5. CONCLUSIONS

Tradeoff between attachment configurations:

- Downforce Maximisation vs Drag Minimisation
- Other concerns: structural/aeroelastic/weight/regulations

Important 3D interference effects:

- Attachment affects flow at the wake and vortical structures
- High level of flow unsteadiness

Computational limitations:

- Memory and time limits for mesh independence



5. CONCLUSIONS

FUTURE LINES OF INQUIRY

Numerical improvement

- Mesh convergence study
- Higher Fidelity: moving road and wheels, URANS

Validation

- Available WT data for Baseline
- New WT tests for attachments

Extension

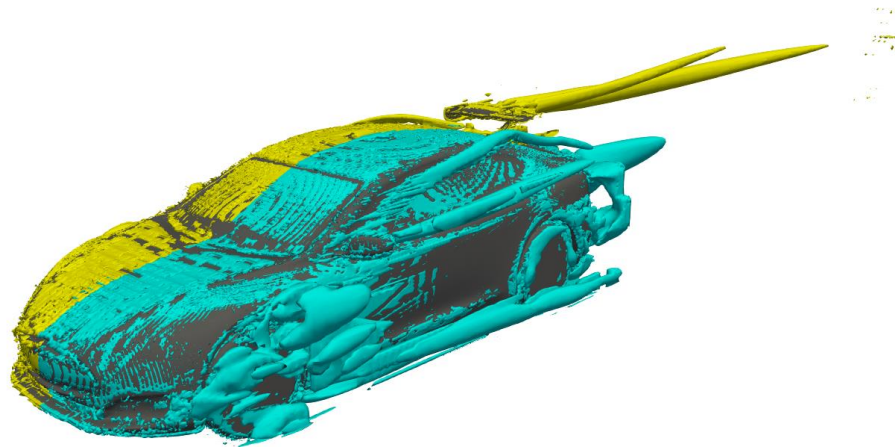
- Sideslip and stability
- Different rear wing geometries and placement





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THANKS FOR YOUR ATTENTION



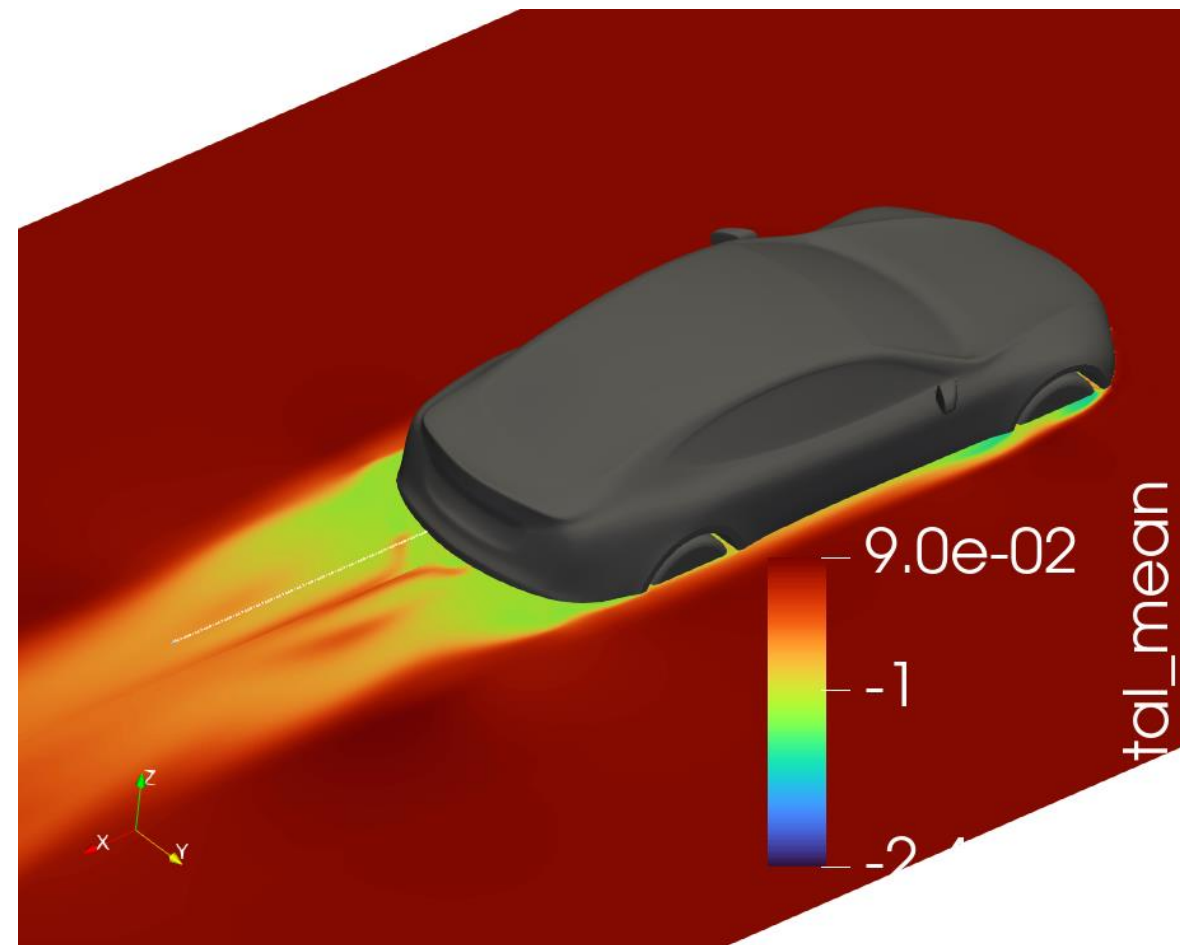
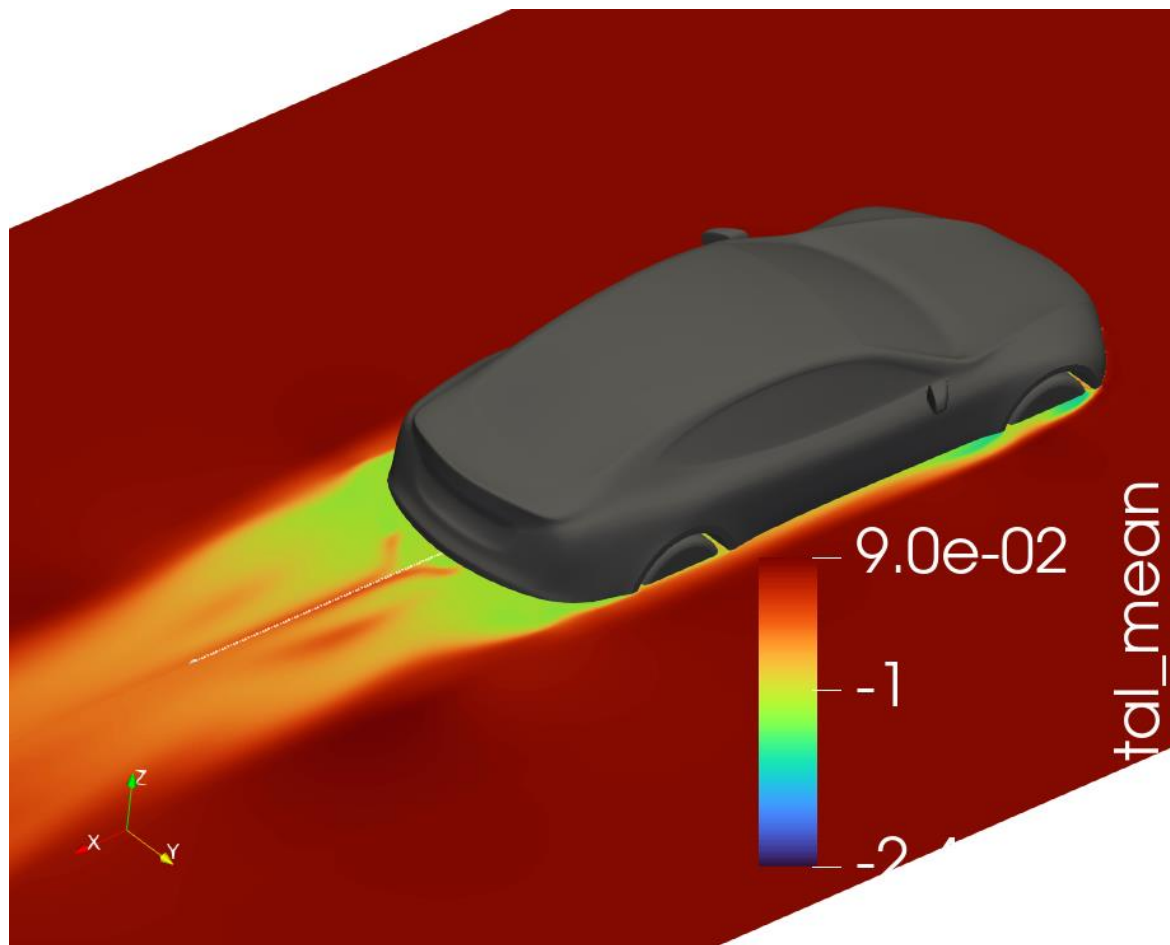
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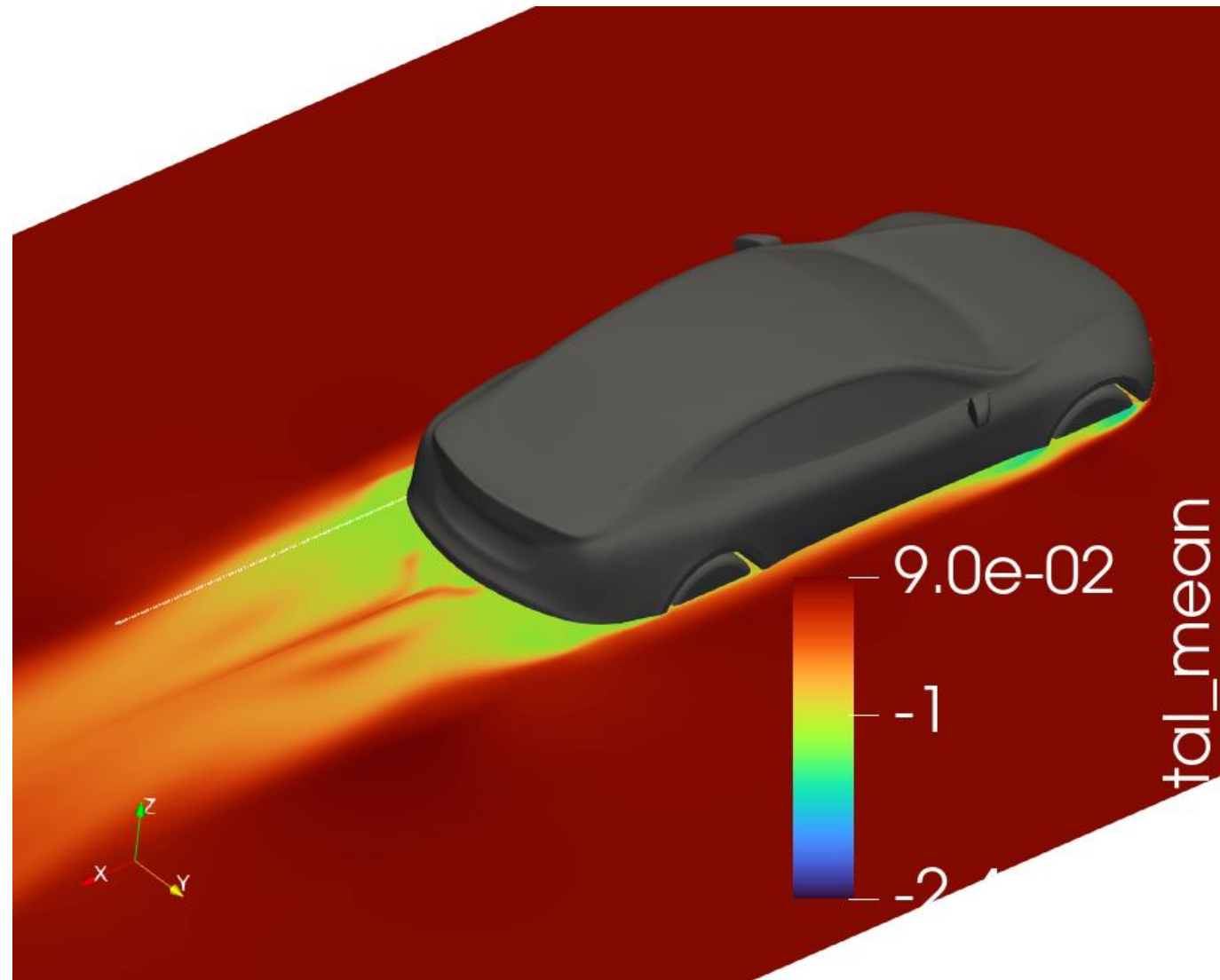
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6. BACK-UP



6. BACK-UP



6. BACK-UP

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